The product of $\boxed{\text{three integers}}$ X, Y and Z is 192. Z equal to 4 and P is equal to the average of X and Y. What is the minimum possible value of P?

(a) 8 (b) 7

(c) 9.5 (d) 6

① $XYZ = 192$

② $Z = 4$

$\Rightarrow XY(4) = 192$

$$\boxed{XY = 48}$$

③ $P = \frac{X+Y}{2}$

$$P_{min} = \frac{(X+Y)_{min}}{2}$$

$$= \frac{14}{2} = 7$$

$XY = 48$

Add^n $(x+y)$

48:

$1 \times 48 \longrightarrow 49$

$2 \times 24 \longrightarrow 26$

$3 \times 16 \longrightarrow 19$

$4 \times 12 \longrightarrow 16$

$6 \times 8 \longrightarrow 14$

If $|9y-6|=3$, then $y^2-(4y/3)$ is

(A) 0

(B) 1/3

(C) -1/3

(D) undefined

[GATE 2016- EE]

$$(9y-6)^2 = 3^2$$

$$81y^2 - 12\times 9y + 36 = 9$$

$$y^2 - \frac{4}{3}y = \frac{(9-36)}{81}$$

M1

$$\boxed{\therefore y^2 - \frac{4}{3}y = -\frac{1}{3}}$$

If $|9y-6|=3$, then $y^2-(4y/3)$ is

(A) 0

(B) 1/3

(C) -1/3

(D) undefined

[GATE 2016- EE]

M2

[Substitution]

$\boxed{y=1}$

- $|9\times1-6| = |9-6| = |3| = 3$
- $y^2-\frac{4y}{3} = 1^2-\frac{4}{3} = -\frac{1}{3}$ //

$\underbrace{a + a + a + \ldots + a}_{n \text{ times}} = a^2b$ & $\underbrace{b + b + b + \ldots b}_{m \text{ times}} = ab^2$, where a, b, n and m are natural numbers. What is the value of

$$\left(\underbrace{m + m + m + \ldots + m}_{n \text{ times}}\right)\left(\underbrace{n + n + n + \ldots + n}_{m \text{ times}}\right)?$$

(a) $2a^2b^2$

(b) a^4b^4

(c) $ab\,(a + b)$

(d) $a^2 + b^2$

[GATE–2018 (CE Afternoon Session)]

$\therefore \quad 1+1+1+1+1 = 5(1)$

$$1+1+1+1 \cdots\cdots n \text{ times} = n(1)$$

$$\Big(\underbrace{m+m+m+\ldots+m}_{n \text{ times}}\Big)\Big(\underbrace{n+n+n+\ldots+n}_{m \text{ times}}\Big)?$$

$$n(m) \qquad m(n)$$

$$= m^2n^2 = a^2b^2 \cdot a^2b^2 = a^4b^4 \;//$$

$\because \boxed{\underbrace{a+a+a+\ldots+a}_{n \text{ times}} = a^2b}$ & $\boxed{\underbrace{b+b+b+\ldots b}_{m \text{ times}} = ab^2}$, where

$n(a) = a^2b$

$m(b) = ab^2$

$\boxed{n = ab}$

$\boxed{m = ab}$

For what values of k given below is $\frac{(k+2)^2}{K-3}$ an integer?

(a) 4, 8, 18
(b) ~~4, 10, 16~~
(c) 4, 8, 28
(d) ~~8, 26, 28~~

[GATE 2018 (EE)]

$k=4$: $\frac{(4+2)^2}{4-3} = \frac{36}{+1} = 36$

$k=8$: $\frac{(8+2)^2}{8-3} = \frac{10^2}{5} = \frac{100}{5} = 20$

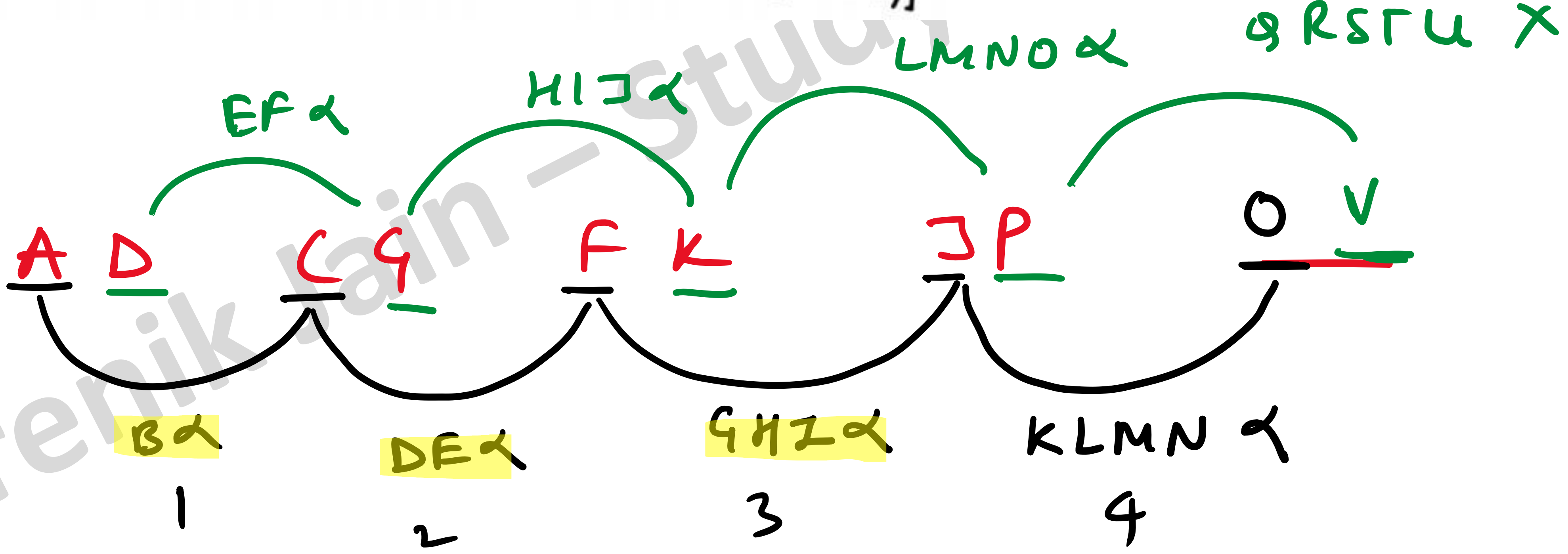
$\cdot\ k = 28 \rightarrow \dfrac{(28+2)^2}{28-3} = \dfrac{30^2}{25} = 36$

Given the sequence of terms, AD CG FK JP, the next ~~time~~ term is

(a) QV (b) OW

(c) PV (d) PW

[CE, ME, CS 2012, 2 Marks (Set-1)]



Find the odd one from the following group:

WEKO IQWA FNTX NVBD

(a) WEKO (b) IQWA

(c) FNTX (d) NVBD

[EC, ME 2014, 2 Marks (Set-1)]

A B C D E F G H I J K L

1 2 3 4 5 6 7 8 9 10 11 12

M N O P Q R S T U V W X Y Z

13 14 15 16 17 18 19 20 21 22 23 24 25 26

$\boxed{W \quad E \quad K \quad O}$

7 5 3

No of alphabets in between

A	B	C	D	E	F	G	H	I	~~J~~	~~K~~	~~L~~
1	2	3	4	5	6	7	8	9	10	11	12

~~M~~	~~N~~	~~O~~	~~P~~	Q	~~R~~	~~S~~	~~T~~	~~U~~	~~V~~	W	~~X~~	~~Y~~	~~Z~~
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$$\boxed{\underbrace{I \quad Q}_{7}\!\!\underbrace{\quad W}_{5}\!\!\underbrace{\quad A}_{3}}$$

No of alphabets in between

A B C D E F ~~G~~ ~~H~~ ~~I~~ ~~J~~ ~~K~~ ~~L~~

1 2 3 4 5 6 7 8 9 10 11 12

~~M~~ N ~~O~~ ~~P~~ ~~Q~~ ~~R~~ ~~S~~ T ~~U~~ ~~V~~ ~~W~~ X Y Z

13 14 15 16 17 18 19 20 21 22 23 24 25 26

$\boxed{F \quad N \quad T \quad X}$

7 5 3

No of alphabets in between

~~A~~	B	~~C~~	D	E	F	G	H	I	J	K	L		
1	2	3	4	5	6	7	8	9	10	11	12		
M	N	~~O~~	~~P~~	~~Q~~	~~R~~	~~S~~	~~T~~	~~U~~	V	~~W~~	~~X~~	~~Y~~	~~Z~~
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$\boxed{N \quad V \quad B \quad D}$

7 5 1

No of alphabets in between

(D) [odd man out]

Find the odd one in the following group

QWZB, BHKM, WCGJ, MSVX,

(a) QWZB (b) BHKM

(c) WCGJ (d) MSVX

[EC, ME 2014, 2 Marks (Set-2)]

~~A~~	B	C	D	E	F	G	H	I	J	K	L		
1	2	3	4	5	6	7	8	9	10	11	12		
M	N	O	P	Q	~~R~~	~~S~~	~~T~~	~~U~~	~~V~~	W	~~X~~	~~Y~~	Z
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$$\boxed{Q \quad W \quad Z \quad B}$$

5 2 1

No of alphabets in between

A	B	~~C~~	~~D~~	~~E~~	~~F~~	~~G~~	H	~~I~~	~~J~~	K	~~L~~
1	2	3	4	5	6	7	8	9	10	11	12

M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$\boxed{B\ \ H\ \ K\ \ M}$

5 2 1

No of alphabets in between

A	B	C	D	E	F	G	H	I	J	K	L
1	2	3	4	5	6	7	8	9	10	11	12

M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$\boxed{W \quad C \quad G \quad J}$

5 3 2

No of alphabets in between

(c) [ODD man out]

A	B	C	D	E	F	G	H	I	J	K	L
1	2	3	4	5	6	7	8	9	10	11	12

M	~~N~~	~~O~~	~~P~~	~~Q~~	~~R~~	S	~~T~~	~~U~~	V	~~W~~	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25	26

$\boxed{M \quad S \quad V \quad X}$

5 2 1

No of alphabets in between

Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

(a) $2^{n-m} P$ (b) $2^{m-n} P$

(c) $2(n-m)P$ (d) $2(m-n)P$

[EC, ME 2014, 1 Mark (Set-4)]

$$\frac{a^m}{a^n} = a^{m-n}$$

$$P = x^n \cdot y^m$$

$$\Rightarrow (2x)^n \left(\frac{y}{2}\right)^m$$

$$\Rightarrow 2^n x^n \cdot y^m \cdot \frac{1}{2^m} = 2^{n-m} P$$

If $\log_x (5/7) = -1/3$, then the value of x is

(a) 343/125

(b) 125/343

(c) –25/49

(d) –49/25

[EC 2015, 1 Mark (Set-1)]

$$\log_x (5/7) = -1/3$$

$$x$$

$$\ln P$$

$$a \qquad P$$

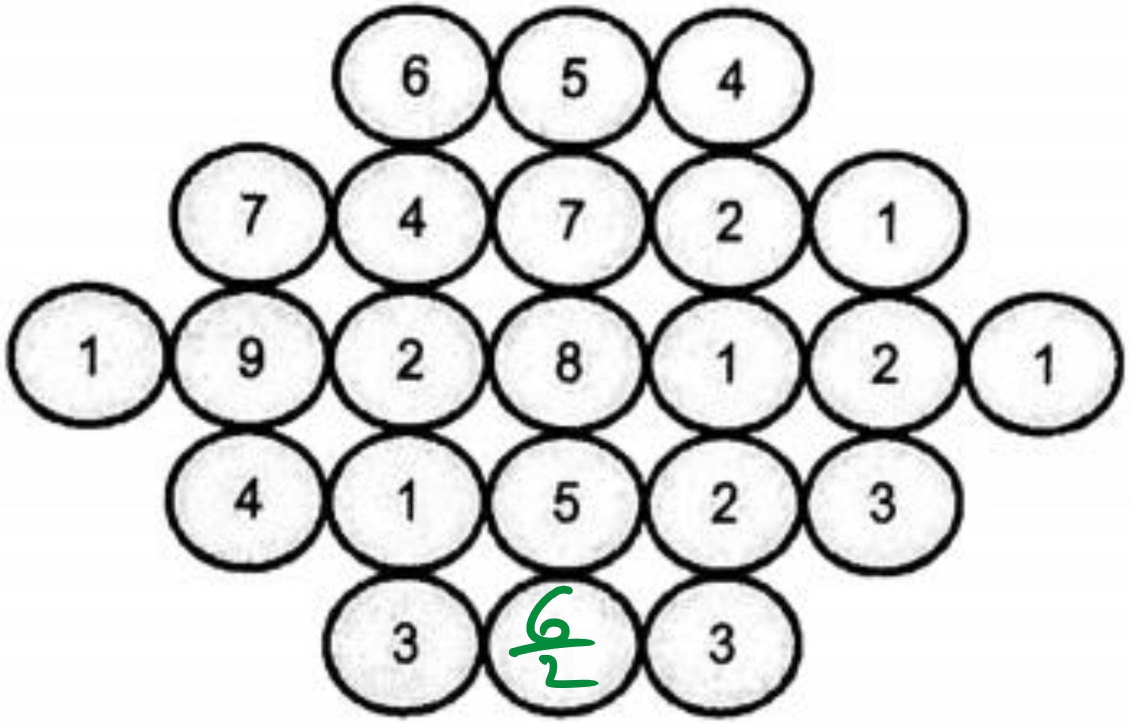
$$\Rightarrow \frac{5}{7} = x^{-1/3} = \frac{1}{x^{1/3}}$$

$$\Rightarrow \quad x^{1/3} = 7/5$$

$$x^{1/3} = 7/5$$

$$x = \frac{7^3}{5^3}$$

Fill in the missing value



=3

[EC 2015, 2 Marks (Set-1)]

log tan1° + log tan2° +......+ log tan89° is ______.

(a) 1

(b) $\frac{1}{\sqrt{2}}$

(c) 0

(d) –1

[EC (Set-3), ME (Set-2) 2015, 2 Marks]

$\ln a + \ln b + \ln c + \ldots = \ln(abc\ldots)$

$\cot 1°$

$\ln[\cancel{\tan 1°} \cdot \cancel{\tan 2°} \cdot \cancel{\tan 3°} \cdots \cancel{\tan 89°}]$

$= \ln[1] = 0$

$\tan(90-\theta) = \cot\theta$

$\tan 89^\circ = \tan(90-1) = \cot 1^\circ$

$\tan\theta = \frac{1}{\cot\theta}$

$\Rightarrow \tan\theta \cdot \cot\theta = 1$

The number that least fits this set:

(324, 441, 97 and 64) is ________.

(a) 324 (b) 441

(c) 97 (d) 64

[EC (Set-3) & IN 2016, 1 Mark]

18^2 21^2 8^2

Pick the odd one out in the following:

13, 23, 33, 43, 53

(a) 23

(b) 33

(c) 43

(d) 53

[EE 2016, 1 Mark (Set-2)]

Composite no.

The last digit of $(2171)^7 + (2172)^9 + (2173)^{11} + (2174)^{13}$ is

(a) 2

(b) 4

(c) 6

(d) 8

[CE 2017, 2 Marks (Set-1)]

$1^7 + 2^9 + 3^{11} + 4^{13}$

$1 + 2^1 + 3^3 + 4$

$1+2+7+4 = 14 = 4$

$$\begin{array}{r} 2 \\ 4\,\overline{)\,9} \\ -8 \\ \hline 1 \end{array} \qquad \begin{array}{r} 2 \\ 4\,\overline{)\,11} \\ -8 \\ \hline 3 \end{array}$$

What is the sum of the missing digits in the subtraction problem below?

$$
\begin{array}{r}
5\_\,\_\,\_\,\_ \\
-48\_89 \\
\hline
1111
\end{array}
$$

(a) 8

(b) 10

(c) 11

(d) Cannot be determined

[ME, 2017, 2 Marks (Set-1)]

$$
\begin{array}{cccccc}
 & 5 & 0 & 0 & 0 & 0 \\
- & 4 & 8 & 8 & 8 & 9 \\
\hline
 & 0 & 1 & 1 & 1 & 1 \\
\hline
\end{array}
$$

Sum of missing digit = 8

$$
\begin{array}{rccccc}
 & 5 & 0 & 1 & 0 & 0 \\
- & 4 & 8 & 9 & 8 & 9 \\
\hline
 & 0 & 1 & 1 & 1 & 1 \\
\hline
\end{array}
$$

Sum $= 0+1+0+0+9$

$= 10$

If a and b are integers and $a - b$ is even, which of the following must always be even?

(a) ab
(b) $a^2 + b^2 + 1$
(c) $a^2 + b + 1$
(d) $ab - b$ → $15 - 3 = 12$ even

[ME, 2017, 1 Mark (Set-2)]

① $a - b$ = even

② $a = 5, b = 3$ → $a - b = 5 - 3 = 2$

Ⓐ $ab = 5 \times 3 \neq$ even

Ⓑ $a^2 + b^2 + 1 = 25 + 9 + 1 \neq$ even

Ⓒ $a^2 + b + 1 = 25 + 3 + 1 = 29 \neq$ even

$o \times e = e$

$o \times o = o$

$e \times o = e$

$e \times e = e$

. Consider a sequence of number a_1, a_2, a_3, ...,

a_n where $\boxed{a_n = \frac{1}{n} - \frac{1}{n+2}}$, for each integer $n > 0$.

What is the sum of the first 50 terms?

(a) $\left(1+\frac{1}{2}\right)-\frac{1}{50}$

(b) $\left(1+\frac{1}{2}\right)+\frac{1}{50}$

(c) $\left(1+\frac{1}{2}\right)-\left(\frac{1}{51}+\frac{1}{52}\right)$

(d) $1-\left(\frac{1}{51}+\frac{1}{52}\right)$

[CE, 2018, 2 Marks (Set-1)]

① $a_1, a_2, a_3 \ldots\ldots$

② $S_{50} = a_1 + a_2 + a_3 + \cdots\cdots a_{50}$

$$= \left(1 - \cancel{\tfrac{1}{3}}\right) + \left(\tfrac{1}{2} - \cancel{\tfrac{1}{4}}\right) + \left(\cancel{\tfrac{1}{3}} - \cancel{\tfrac{1}{5}}\right)$$

$$+ \left(\cancel{\tfrac{1}{4}} - \cancel{\tfrac{1}{6}}\right) + \left(\cancel{\tfrac{1}{5}} - \tfrac{1}{7}\right) + \left(\cancel{\tfrac{1}{6}} - \tfrac{1}{8}\right)$$

$$+ \left(\tfrac{1}{46} - \cancel{\tfrac{1}{48}}\right)$$

$$\left(\tfrac{1}{47} - \cancel{\tfrac{1}{49}}\right) + \left(\cancel{\tfrac{1}{48}} - \cancel{\tfrac{1}{50}}\right) + \left(\cancel{\tfrac{1}{49}} - \tfrac{1}{51}\right) + \left(\cancel{\tfrac{1}{50}} - \tfrac{1}{52}\right)$$

What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, ___x___, 0

(a) 2880 ~~(b) 1440~~

(c) 720 (d) 0

[CS, 2018, 1 Mark]

$\frac{x}{1440} = 1$

$x = 1440$

$\frac{12}{2} = 6$

$\frac{60}{12} = 5$

$\frac{240}{60} = 4$

Given that a and b are integers and $a + a^2b^3$ is odd, which one of the following statements is correct?

(a) a and b are both odd

(b) a ~~and b~~ are both even

(c) ~~a is~~ even and b is odd

(d) a is odd and b is even

[ME, 2018, 2 Marks (Set-1)]

$\boxed{\text{odd}}$

$a\,[\,1 + \underbrace{ab^3}_{\text{odd}}\,] = \text{odd}$

even

$ab^3 = \text{even}$

odd even